

CHAPTER 9

UNIFORM ELECTRONIC SEMICONDUCTORS IN EQUILIBRIUM

9.1 SEMICONDUCTORS

In the preceding chapters our main concern has been to understand the electrical and thermal properties of metallic conductors and of insulators, and to understand why some substances are extremely good electrical conductors and others are extremely poor ones. These objectives have now been accomplished in a general way, and from this point on we shall turn our attention toward a rather detailed examination of the properties of semiconductors. There are three reasons why this is a useful thing to do. First, it is usually possible to use Maxwell-Boltzmann statistics in discussing the statistical mechanics of charge carriers in semiconductors. This means that it is possible to obtain exact analytical solutions for many problems in semiconductors which can be solved only by approximate or numerical methods for metals, where Fermi-Dirac statistics must be used. An understanding of such effects in semiconductors may furnish valuable physical insight into their counterparts in metallic substances. In short, it is possible to understand electronic processes more easily and more deeply in semiconductors than in any other class of crystals. Second, the technology of crystal growth has made it possible to produce crystals of certain semiconducting substances (notably germanium and silicon) of fantastic purity and crystalline perfection, far in excess of what can be obtained at present for metals and insulators. The existence of these nearly perfect crystals makes it possible for the experimentalist to observe electronic transport properties and thermoelectric and galvanomagnetic effects with ease and precision, secure in the knowledge that what he is trying to observe will not be obscured by effects caused by impurities or structural imperfections in the crystal lattice. We may thus easily observe and interpret phenomena in semiconductors which would be difficult or impossible either to measure accurately or to explain quantitatively in metallic substances. Finally, semiconductors have acquired over the last two decades an enormous technological importance. They have come to be used in making all sorts of electronic devices, including rectifiers, transistors, photocells, voltage regulators, parametric amplifiers, and switching devices. An appreciation of the fundamental electronic transport properties in these substances is of direct application in understanding and analyzing the operation of all the device structures of contemporary technical importance.

A semiconductor is a crystalline substance having an energy band structure in which a band of electronic states, completely filled at zero temperature, is separated from one which is completely empty at absolute zero by a narrow region of forbidden

energies. This band structure is shown schematically in Figure 9.1(a). At absolute zero the semiconductor is a perfect insulator, since there are no partially filled bands. At higher temperatures, however, a few electrons from the valence band may acquire enough random thermal energy to be excited across the forbidden band to become conduction electrons in the heretofore empty *conduction* band. The empty states left behind in the lower or *valence* band also contribute to the conductivity, behaving like

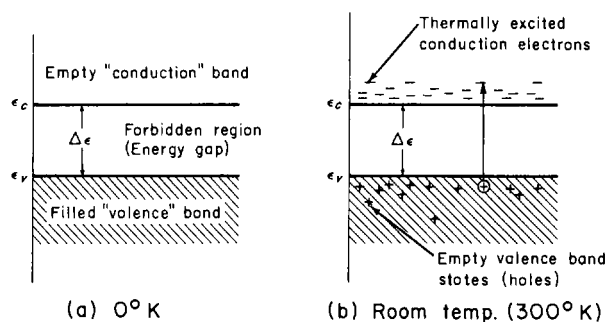


FIGURE 9.1. Conduction and valence bands of a pure semiconductor (a) at absolute zero, (b) at room temperature, showing thermally excited electrons and holes.

positively charged holes. It is clear that the number of conduction electrons and the number of holes must rise with increasing temperature, and hence the electrical conductivity likewise increases with temperature.

The physical mechanism of electron and hole conduction in *covalent* semiconductors such as carbon (diamond), germanium, and silicon, which form crystals having the diamond structure of Figure 1.9 can be understood from Figure 9.2. This figure shows the calculated energy band structure of diamond, plotted against the interatomic spacing, as in Figure 8.10. The corresponding diagrams for silicon and germanium are quite similar. When N isolated atoms are assembled into a crystal, the $2s$ and $2p$ atomic levels first broaden into energy bands; as the interatomic spacing decreases these bands broaden further, and eventually overlap. As the interatomic distance becomes still smaller, the continuum of what were originally $2s$ and $2p$ states splits once more into two bands, each of which, however, now contain precisely $4N$ states. At the equilibrium interatomic distance a , these bands are separated by an "energy gap" or forbidden region of width $\Delta\epsilon$. Since the electronic structure of carbon is $1s^2)2s^22p^2$, there are $4N$ valence electrons available, which exactly suffice to fill the lower of the two bands, and form the valence band of the crystal. The upper band becomes the conduction band. The forbidden energy region under normal conditions has a width of about 7 eV in diamond, 1.2 eV in silicon and 0.7 eV in germanium. Due to thermal expansion of the lattice, this "energy gap" has a weak dependence on temperature; it is clear from Figure 9.2 that the energy gap will decrease as the crystal expands. It is also evident from this diagram that $\Delta\epsilon$ will be a function of pressure, becoming larger as the interatomic spacing is reduced by the application of pressure by hydrostatic or other means. Beyond the point A in Figure 9.2, where the $2p$ and $2s$ bands overlap, the s and p character of the electronic states is lost; in the valence

band of the semiconductor, therefore, the electron wave functions are a mixture of s and p atomic wave functions.

The electrons in the valence band are the electrons which form the tetrahedrally disposed electron-pair covalent bonds between the atoms in Figure 1.9. The thermal excitation of an electron from the valence band to the conduction band corresponds physically to the removal of an electron from a covalent pair bond by thermal agitation

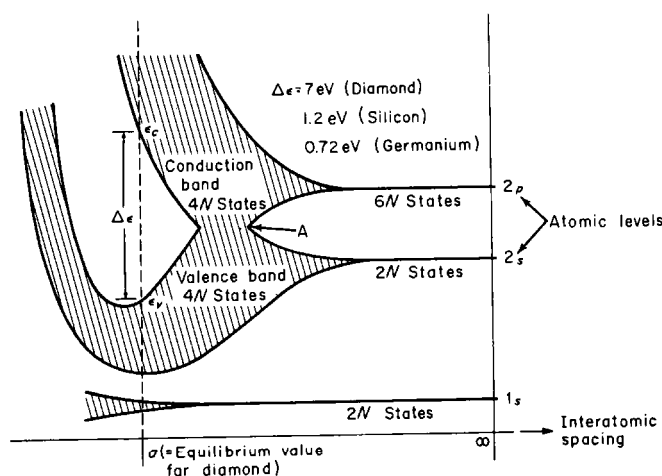


FIGURE 9.2. The bands arising from the $1s$, $2s$, and $2p$ atomic states of carbon (diamond) as a function of interatomic distance. [After W. Shockley, *Electrons and Holes in Semiconductors*, D. van Nostrand Co., Inc., New York (1950).]

of the lattice. The electron then becomes a free electron, outside the covalent lattice bond scheme, and is available to function as a charge carrier to conduct current through the crystal. This process is depicted schematically in Figure 9.3(a), though it must be remembered that the bond scheme shown there is a two-dimensional rendering of what in actuality is a three-dimensional network of tetrahedral bonds. The excitation of an electron leaves behind a localized defect in the covalent bond structure of the crystal, which can be identified as a valence band state which could be occupied by an electron but is actually empty. This defect is simply the "hole" of Section 8.5. Both the free electron and the hole are migratory; the free electron may wander about within the crystal in a random fashion, impelled by thermal energy it may acquire from the lattice. The hole, likewise, may migrate by virtue of the fact that an electron in a covalent bond pair adjacent to the hole may very easily move into the hole, completing the bond pair at the former hole site, but transferring the location of the hole to the site whence the electron came. This hole migration may also be caused by thermal agitation of the lattice. Free electrons and holes will also move in response to an electric field, and may then give rise to a macroscopic current flow through the crystal. This situation is illustrated in Figure 9.3(b). Here *all* the electrons in the crystal, in both valence band and conduction band, are subjected to a force $-eE$, acting to the right in the diagram. The free electrons move to the right, causing a

conventional current flow to the left in view of their negative charge. In addition to this, an electron in a covalent bond pair adjacent to a hole may move to the right into the empty electron site associated therewith, the hole then moving to the left, to the site whence this electron came; this process may now be repeated, the net result being a transfer of an electron to the right, accompanied by the motion of the hole to the

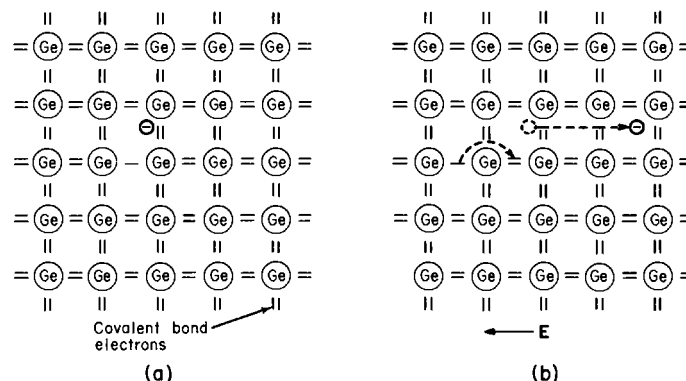


FIGURE 9.3. (a) A free electron and a hole arising from thermal ionization of an electron originally in a covalent bond. (b) Motion of free electron (to the right) and hole (to the left) when an electric field is applied as shown.

left, hence in the direction in which a *positively* charged particle would move under the influence of the applied field. The net electron current to the right again causes a conventional current flow to the left, which might as well be represented as a current of *positive holes* in that direction. Current flow may thus arise both from the movement of free electrons and from the migration of valence electrons into and out of empty states in the valence band, which we prefer to think of as the migration of positive holes.

9.2 INTRINSIC SEMICONDUCTORS AND IMPURITY SEMICONDUCTORS

A semiconductor in which holes and electrons are created solely by thermal excitation across the energy gap is called an *intrinsic* semiconductor. Holes and electrons which are created in this manner are often referred to as *intrinsic charge carriers* and the conductivity arising from such carriers is sometimes termed *intrinsic conductivity*. In an intrinsic semiconductor the concentrations of electrons and holes must always be *the same*, since the thermal excitation of an electron inevitably creates one and only one hole.

The population of holes and electrons in an intrinsic semiconductor is described statistically in terms of the Fermi-Dirac distribution function and the density of states functions for the valence and conduction bands. Since the bottom of the conduction

band and the top of the valence band exhibit an essentially parabolic dependence of ϵ on k , the behavior of electrons and holes in these regions is essentially that of free particles, with appropriate effective mass factors. Electrons and holes are only very rarely excited into regions of the conduction and valence bands where their properties may depart from free-particle behavior at physically realizable temperatures, and so the effects of such excitations may for practical purposes be neglected. The density of states functions to be used, then, are essentially those for free particles, the density of states in the conduction band being given by

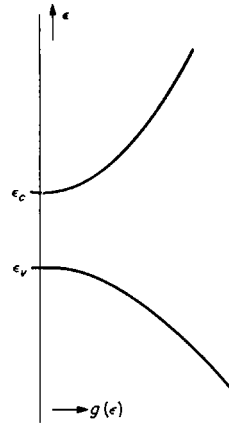
$$g_c(\epsilon) d\epsilon = \frac{8\sqrt{2}\pi}{h^3} m_n^{*3/2} \sqrt{\epsilon - \epsilon_c} d\epsilon \quad (\epsilon > \epsilon_c) \quad (9.2-1)$$

and the density of states in the valence band by

$$g_v(\epsilon) d\epsilon = \frac{8\sqrt{2}\pi}{h^3} m_p^{*3/2} \sqrt{\epsilon_v - \epsilon} d\epsilon, \quad (\epsilon < \epsilon_v) \quad (9.2-2)$$

where m_n^* is the effective mass for electrons in the conduction band, and m_p^* the effective mass for holes in the valence band.¹ The density of states in the forbidden region $\epsilon_v < \epsilon < \epsilon_c$ is, of course, zero. A plot of the density of states curve for an intrinsic semiconductor is shown in Figure 9.4. If m_p^* and m_n^* are precisely equal, the

FIGURE 9.4. Density of states function for an intrinsic semiconductor.



Fermi energy must lie exactly in the center of the forbidden region. This is true because otherwise the population of electrons in the conduction band and holes in the valence band, obtained by integrating the product of the density of states function and the probability factor $f_0(\epsilon)$ for electrons in the conduction band or $1 - f_0(\epsilon)$ for holes in the valence band would not be equal. The situation is represented diagrammatically in Figure 9.5(a). If m_p^* and m_n^* are not equal (and this is usually the case), the

¹ The subscripts n and p refer to *negative* and *positive* charge carriers. This convention will be used consistently.

Fermi energy must be adjusted upwards or downwards a bit, away from the precise center of the energy gap, to equalize the population integrals, and hence must lie *near* but not *at* the center of the forbidden region. This state of affairs is shown in Figure 9.5(b). We shall investigate the statistics of holes and electrons in some detail in a later section.

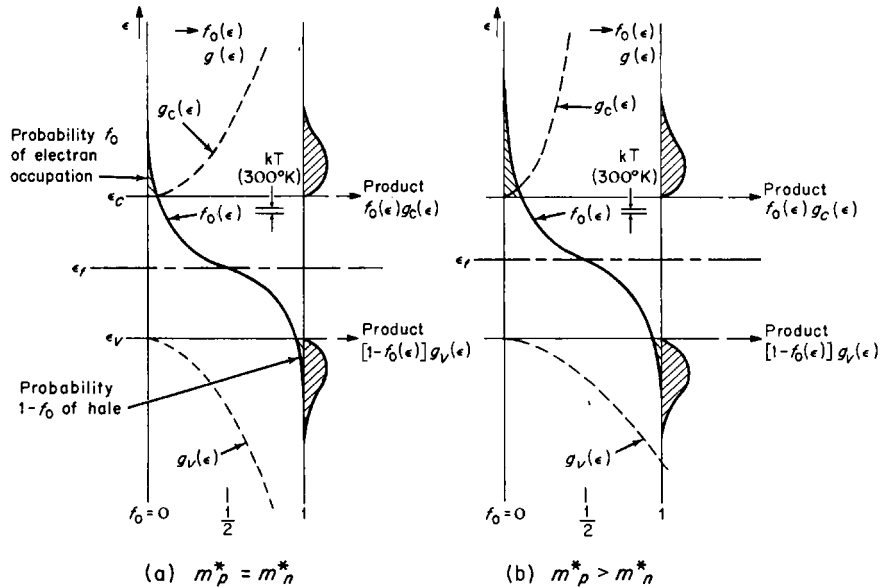


FIGURE 9.5. Distribution function, Fermi level, density of states function and electron and hole populations for an intrinsic semiconductor (a) where $m_p^* = m_n^*$ and (b) where $m_p^* > m_n^*$. The "spread" of the Fermi distribution is exaggerated here for the sake of illustrative clarity; in reality at 300°K the Fermi distribution would be much more like the "step" function which is found for $T = 0$.

It is quite easy to introduce very small amounts of substances such as arsenic, antimony or other elements belonging to group V of the periodic table into otherwise pure crystals of silicon or germanium as substitutional impurities, that is, as impurity atoms which occupy lattice sites which would normally be occupied by atoms of the covalent semiconductor. The group V atoms have five valence electrons apiece. Four of these are used in forming covalent electron pair bonds with neighboring semiconductor atoms; the fifth is bound to the impurity atom only by electrostatic forces which are quite weak, and it can therefore be easily ionized by thermal agitation of the lattice at ordinary temperatures to provide an *extra* conduction electron. The impurity atom which is left behind then becomes a positive ion, which, however, is *immobile* in view of the fact that it is strongly bound to four neighboring atoms by the usual covalent bonds. The situation is illustrated in Figure 9.6(a). In crystals containing this type of impurity there are more electrons than holes (although *some* holes are still present because hole-electron pairs are still created thermally on occasion). Such crystals are termed *n-type* semiconductors, the terminology arising from the fact that

most of the charge carriers are *negative* electrons. The component of electrical conductivity arising from the impurity atoms is called *impurity conductivity*, and a substance most of whose charge carriers originate from impurity atoms is called an *impurity semiconductor*. The substitutional group V atoms are often called *donor* atoms, because they each donate an additional free electron to the crystal. It takes so little energy (as we shall soon see) to ionize a typical group V donor impurity atom in Si and Ge that essentially *all* group V impurities in these materials are ionized at temperatures above about 20°K.

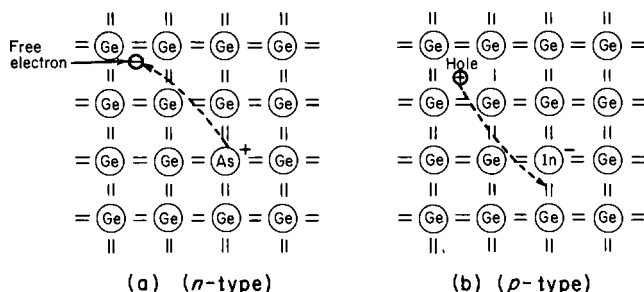


FIGURE 9.6. (a) Free electron arising from ionization of a substitutional arsenic impurity atom; (b) free hole arising from ionization of a substitutional indium impurity atom.

If, instead of group V atoms, group III impurity atoms (Al, Ga, In, and so forth) are introduced substitutionally into the lattice, a rather different effect is observed. These atoms have only three valence electrons, which are used in forming covalent pair bonds to three neighboring atoms, but the fourth bond of necessity lacks an electron. There is, in effect, an extra *hole* built into the covalent bond structure at the impurity atom. This hole can easily migrate away from the impurity site, because an extra electron from a neighboring covalent bond can migrate to the impurity site and fill in the fourth electron pair bond there (which, of course, puts a *negative* charge on the impurity atom); the hole is then associated with a neighboring atom, indistinguishable from a thermally created hole. This state of affairs is illustrated in Figure 9.6(b). The energy required for the migration of the hole away from the impurity site is of the order of the energy required to remove the extra electron from a donor atom. Therefore, except at very low temperatures, all the holes will be migratory and all the group III impurity atoms will have the character of immobile negative ions. In crystals containing predominantly this type of impurity there are more holes than electrons, although there will always be some electrons arising from thermal excitation. Crystals of this sort are called *p-type* semiconductors, because the charge carriers are for the most part *positive*. The substitutional group III atoms are usually termed *acceptor* atoms, because they readily accept an electron from the covalent bond structure, freeing a mobile hole. When impurities of either type are present in a semiconductor crystal, the conductivity is invariably *greater* than that associated with a pure or intrinsic semiconductor at the same temperature, due to the extra charge carriers contributed by the impurity atoms, and, in general, the greater the impurity concentration, the greater the conductivity.

The statistical picture of n -type and p -type semiconductors is characterized by the presence of the Fermi level above (for n -type) or below (for p -type) the position associated with the pure or intrinsic crystal. In an n -type crystal, for example, there cannot be more electrons than holes *unless* the Fermi level is adjusted *upward* from the

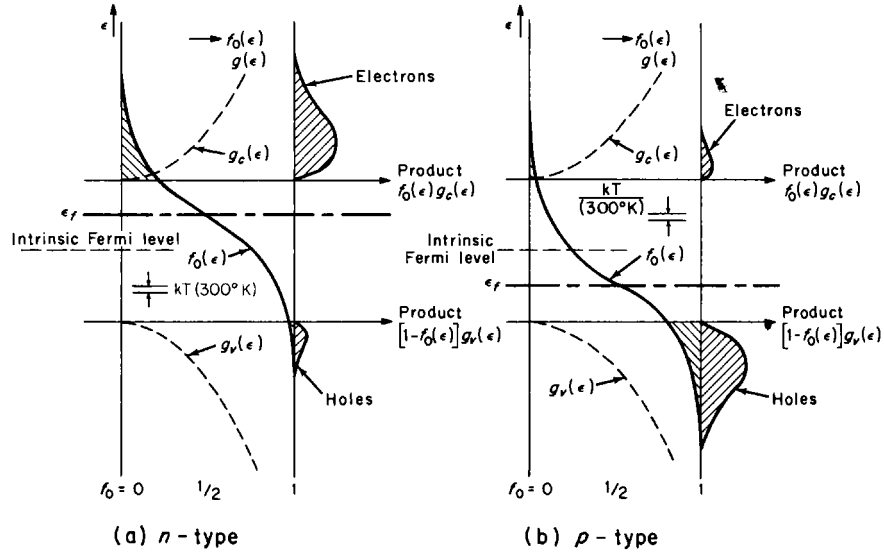


FIGURE 9.7. Distribution function, Fermi level, and electron and hole populations for (a) an n -type impurity semiconductor and (b) a p -type impurity semiconductor. Again, as in Figure 9.5, the “spread” of the Fermi distribution is exaggerated for illustrative purposes.

intrinsic position, and vice-versa for p -type, as shown in Figure 9.7. As the temperature and impurity concentration are varied, the position of the Fermi level changes in a rather complex way, about which we shall have more to say later.

9.3 STATISTICS OF HOLES AND ELECTRONS—THE CASE OF THE INTRINSIC SEMICONDUCTOR

Under conditions of thermal equilibrium, the number of electrons dn_0 per unit volume having energy in a range $d\epsilon$ about ϵ in the conduction band of *any* semiconductor, intrinsic or impurity, is, according to the results of Chapter 5

$$dn_0 = f_0(\epsilon)g_c(\epsilon) d\epsilon = \frac{8\sqrt{2}\pi}{h^3} m_n^{*3/2} \frac{\sqrt{\epsilon - \epsilon_c} d\epsilon}{1 + e^{\frac{\epsilon - \epsilon_f}{kT}}} \quad (9.3-1)$$

where $f_0(\epsilon)$ represents the equilibrium Fermi distribution function (5.5-20) and $g_c(\epsilon)$ is the density of states factor (9.2-1). Now $\Delta\epsilon$, the width of the forbidden energy

region is typically of the order of 1 eV, and kT at room temperature (300°K) is only about 1/40 eV. If the Fermi energy is *within the forbidden energy region*, several kT units away from the edge of the conduction band (that is, if $\epsilon_c - \epsilon_f \gg kT$), then for all energies belonging to the conduction band, the exponential factor in the denominator of the Fermi distribution function is much greater than unity. We may then represent the Fermi distribution function in the conduction band as

$$f_0(\epsilon) \cong e^{-(\epsilon - \epsilon_f)/kT} \quad (9.3-2)$$

which is a distribution function of essentially Maxwellian form. The situation here is substantially that which was discussed in Chapter 5 in connection with Equation (5.5-27). Since we have already shown in Figure 9.5 that the Fermi energy for an intrinsic semiconductor must lie well within the forbidden region of energy, and since $kT \ll \Delta\epsilon$ at all accessible temperatures as pointed out above, the condition $\epsilon_c - \epsilon_f \gg kT$ will always be satisfied in practice, and we shall always be able to use (9.3-2) for intrinsic semiconductors rather than the more complicated expression (5.5-20). As a matter of fact, in most cases involving impurity semiconductors we shall find that this condition is also satisfied, allowing us to use (9.3-2) also for impurity semiconductors. It is clear, for example, that this simplification would be valid for the impurity semiconductors of Figure 9.7. In both Figures 9.5 and 9.7 it is easily seen that the exponential "tail" of the Fermi distribution function is the only part of that function which overlaps the conduction and valence bands. The part of the Fermi function in the neighborhood of the Fermi energy, which is invariably the part which causes mathematical difficulties, coincides in all these cases with a region where the density of available electronic states is *zero*. We shall refer to the simplifying approximation (9.3-2) as the *Boltzmann approximation*.

Using this approximation in (9.3-1) and integrating to get the total number of electrons per unit volume in the conduction band, we find

$$n_0 = \int_{\epsilon_c}^{\infty} dn_0 = \frac{8\sqrt{2}\pi}{h^3} m_n^{*3/2} e^{\epsilon_f/kT} \int_{\epsilon_c}^{\infty} \sqrt{\epsilon - \epsilon_c} e^{-\epsilon/kT} d\epsilon. \quad (9.3-3)$$

Actually, the limits of integration should be from ϵ_c to the top of the conduction band, wherever that may be; however, since $f_0(\epsilon)$ falls off very rapidly with increasing ϵ (at least for reasonable values of temperature) most of the electrons are concentrated in states near the bottom of the conduction band, and so long as the top of the band is many kT units higher in energy than ϵ_c , it makes little difference whether we integrate to the top of the band or to infinity. Making the substitution

$$x_c = \frac{\epsilon - \epsilon_c}{kT} \quad (9.3-4)$$

the integral (9.3-3) can be expressed in the form

$$n_0 = \frac{8\sqrt{2}\pi}{h^3} (m_n^* kT)^{3/2} e^{\frac{\epsilon_f - \epsilon_c}{kT}} \int_0^{\infty} x_c^{1/2} e^{-x_c} dx_c. \quad (9.3-5)$$

According to (5.4-4), the value of the integral above is $\Gamma(3/2) = \sqrt{\pi}/2$. Substituting

this value for the integral, the final result may be written

$$n_0 = U_c e^{-\frac{e_c - e_f}{kT}} \quad (9.3-6)$$

where

$$U_c = 2(2\pi m_n^* kT/h^2)^{3/2}. \quad (9.3-7)$$

The numerical value of the quantity U_c can be written (using cgs units) as

$$U_c = (4.83 \times 10^{15})(m_n^*/m_0)^{3/2} T^{3/2} \text{ cm}^{-3}, \quad (9.3-8)$$

where m_0 is the inertial mass of the electron. For $m_n^* = m_0$ and $T = 300^\circ\text{K}$, this gives $U_c = 2.51 \times 10^{19} \text{ cm}^{-3}$.

The number of holes p_0 per unit volume in the valence band of a semiconductor can be found in a similar manner. In this case we may write as dp_0 , the number of holes in the valence band in a range of energy $d\varepsilon$ about ε , the expression

$$dp_0 = f_{p0}(\varepsilon) g_v(\varepsilon) d\varepsilon = [1 - f_0(\varepsilon)] g_v(\varepsilon) d\varepsilon. \quad (9.3-9)$$

The quantity $f_{p0}(\varepsilon)$ is the probability that a hole will be associated with a quantum state of energy ε , in other words, that such a state will be unoccupied. This probability is just one minus the probability of occupation $f_0(\varepsilon)$. Since $f_0(\varepsilon)$ is given by (5.5-20) it is easily seen that

$$f_{p0}(\varepsilon) = 1 - f_0(\varepsilon) = 1 - \frac{1}{1 + e^{(\varepsilon - e_f)/kT}} = \frac{1}{1 + e^{(e_f - \varepsilon)/kT}}. \quad (9.3-10)$$

Again, if the Fermi energy is several kT units above the edge of the valence band, that is, if $e_f - e_v \gg kT$, the exponential factor in the denominator of the expression (9.3-10) will be much larger than unity for all values of ε in the valence band, whereby one may set

$$f_{p0}(\varepsilon) \cong e^{-(e_f - \varepsilon)/kT}. \quad (9.3-11)$$

This is the Boltzmann approximation for holes in the valence band. Using (9.3-11) and the density of states function (9.2-2) in (9.3-9) and integrating over the valence band we obtain

$$p_0 = \frac{8\sqrt{2}\pi}{h^3} m_p^{*3/2} e^{-e_f/kT} \int_{-\infty}^{e_v} \sqrt{e_v - \varepsilon} e^{\varepsilon/kT} d\varepsilon. \quad (9.3-12)$$

Again, the lower limit of the integral has been taken as $-\infty$ rather than the bottom of the valence band; this simplification can be justified by essentially the same argument which was made in connection with (9.3-3) for the conduction band. Making the substitution

$$x_v = \frac{e_v - \varepsilon}{kT} \quad (9.3-13)$$

the integral can be expressed in the form

$$p_0 = \frac{8\sqrt{2}\pi}{h^3} (m_p^* kT)^{3/2} e^{-(\epsilon_f - \epsilon_v)/kT} \int_0^\infty x_v^{1/2} e^{-x_v} dx_v, \quad (9.3-14)$$

which can be evaluated as a Γ -function as above to give

$$p_0 = U_v e^{-(\epsilon_f - \epsilon_v)/kT} \quad (9.3-15)$$

with

$$U_v = 2(2\pi m_p^* kT/h^2)^{3/2} \quad (9.3-16)$$

The numerical value of U_v may be obtained from (9.3-8), substituting m_p^* in the formula for m_n^* .

The product $n_0 p_0$ can easily be shown to be a function *only* of the energy gap $\Delta\epsilon$, the effective masses, and the temperature, *independent* of the Fermi level or impurity content. Thus, multiplying the expressions (9.3-6) and (9.3-15) for n_0 and p_0 , one may obtain

$$n_0 p_0 = U_c U_v e^{-(\epsilon_c - \epsilon_v)/kT} = U_c U_v e^{-\Delta\epsilon/kT}. \quad (9.3-17)$$

Using (9.3-7) and (9.3-16) this can be written in the form

$$n_0 p_0 = 4 \left(\frac{2\pi(m_p^* m_n^*)^{1/2} kT}{h^2} \right)^3 e^{-\Delta\epsilon/kT}. \quad (9.3-18)$$

For a given semiconducting substance, the effective masses and the energy gap $\Delta\epsilon$ are fixed; hence the product $n_0 p_0$ in a given material must be a function only of temperature. This is essentially a mass-action law governing the relative concentrations of holes and electrons in a given material. If the semiconductor is in the pure or intrinsic state, then the concentrations of holes and electrons must be equal, as we have already seen, because the only holes and electrons which can be present are those which are generated in pairs by direct thermal excitation of valence band electrons. In such a material, then

$$p_0 = n_0 = n_i(T) \quad (9.3-19)$$

where $n_i(T)$ stands simply for the number of holes or electrons per unit volume in an *intrinsic* sample of the semiconductor in question at temperature T . In a sample of the same substance which is *not* intrinsic but rather an *impurity* semiconductor, the number of holes and electrons is no longer equal; nevertheless, according to (9.3-17) *the product $n_0 p_0$ must be the same for this material as for the intrinsic substance.* We may then write, from (9.3-19)

$$n_0 p_0 = n_i^2(T) \quad (9.3-20)$$

with

$$n_i(T) = 2 \left(\frac{2\pi(m_p^* m_n^*)^{1/2} kT}{h^2} \right)^{3/2} e^{-\Delta\epsilon/2kT}, \quad (9.3-21)$$

as given by (9.3-18). According to these results, the concentration of carriers in an intrinsic semiconductor is a strong function of temperature, increasing rapidly as the temperature rises, and is likewise strongly dependent upon the energy gap $\Delta\epsilon$, decreasing rapidly as $\Delta\epsilon$ increases.

The mass action relation (9.3-17) or (9.3-20) is strongly reminiscent of the laws governing the relative concentrations of ions in substances which are weakly ionized, for example the H^+ and OH^- ions in aqueous systems. As a matter of fact, semiconductors can be understood as an example of the classical Arrhenius theory of weakly ionized electrolytes in terms of a system in which a covalent bond dissociates into an electron and a hole with dissociation energy $\Delta\epsilon$.

In a purely intrinsic sample, where the hole and electron concentrations are equal, we may equate expressions (9.3-6) and (9.3-15), take the logarithm of both sides of the resulting equation, and solve for the Fermi energy, obtaining

$$\epsilon_f = \frac{1}{2}(\epsilon_v + \epsilon_c) + kT \ln \sqrt{U_v/U_c} \quad (\text{intrinsic}). \quad (9.3-22)$$

Since U_v and U_c are given by (9.3-16) and (9.3-7), we may express this result in the form

$$\epsilon_{fi} = \frac{1}{2}(\epsilon_v + \epsilon_c) + kT \ln(m_p^*/m_n^*)^{3/4} \quad (\text{intrinsic}), \quad (9.3-23)$$

using the symbol ϵ_{fi} to denote the Fermi energy of an intrinsic semiconductor. It is clear from this that in an intrinsic semiconductor the Fermi energy is displaced from the center of the energy gap by only a relatively small amount of energy as given by the second term on the right-hand side of (9.3-23). If $m_p^* = m_n^*$, the Fermi level is exactly at the midpoint of the forbidden region, as indicated previously.

9.4 IONIZATION ENERGY OF IMPURITY CENTERS

A donor center in a semiconductor consists of a fixed ion of charge $+e$ to which an electron is loosely bound. If the binding energy is small enough, the "orbit" of the electron will be quite large—so large, indeed, compared to the interatomic spacing that many atoms will be encompassed by the path of the electron about the donor ion.² Under these circumstances it will be approximately correct to regard the electron as being immersed in a uniform polarizable medium whose dielectric constant κ is the macroscopic dielectric constant of the semiconductor crystal, since the electrostatic force between electron and donor ion will then be modified, on the average, by the polarization of many intervening atoms. The picture is similar to that of a hydrogen atom in a uniform continuous medium of dielectric constant κ , if one can envision such a situation. This picture will be valid as long as κ is of sufficient magnitude so that the Bohr orbits are large compared with the interatomic spacing.

We may now work out the size of the orbits and associated energy levels according

² Although descriptive of the situation, this is not very good wave mechanical language; it would be more accurate to say that the wave function of the electron extends over many interatomic distances, so that many atoms are included within the region where the probability density is large.

to the Bohr theory discussed in Section 4.5. It would really be more accurate to use the wave mechanics of the hydrogen atom for these calculations, but since we know that the Bohr theory gives the correct energy levels, and since the Bohr radii and the wave mechanical average electron-nuclear distances are almost the same, we are confident that the simpler Bohr theory will yield the right answers for this very similar situation. The quantization condition for the orbital angular momentum holds in both cases, so that Equation (4.5-2) is still correct provided the electron mass is replaced by m_n^* . On the other hand, due to the polarization of the crystal, the electrostatic force between the donor ion and the bound electron is reduced by a factor κ , becoming $e^2/\kappa r_n^2$, so that Equation (4.5-3) should now be written

$$e^2/\kappa r_n^2 = m_n^* v_n^2 / r_n = m_n^* r_n \omega_n^2. \quad (9.4-1)$$

Equations (9.4-1) and (4.5-2) may now be solved as simultaneous equations for r_n and ω_n giving [in analogy to (4.5-4) and (4.5-5)],

$$r_n = \frac{n^2 \hbar^2 \kappa}{m_n^* e^2} \quad (9.4-2)$$

and

$$\omega_n = \frac{m_n^* e^4}{n^3 \hbar^3 \kappa^2} \quad (9.4-3)$$

The kinetic and potential energies may now be evaluated as in (4.5-6) and (4.5-7), recalling, however, that the potential energy is now given by $-e^2/\kappa r_n$, whereby the total energy is

$$\varepsilon_n = \varepsilon_k + \varepsilon_p = -\frac{m_n^* e^4}{2n^2 \kappa^2 \hbar^2}. \quad (9.4-4)$$

The Bohr theory for the hydrogen atom is expressed by the above equations with $m_n^* = m_0$ and $\kappa = 1$. It is evident from (9.4-2) that the Bohr orbits for the donor center are larger than the hydrogen orbits by a factor κ and also by a factor m_0/m_n^* . For n -type germanium, $\kappa = 16$ and $m_n^* \cong m_0/4$, so that the radius of the first Bohr orbit is 64 times the radius of the first Bohr orbit of hydrogen. The value for hydrogen is 0.528 Å, so that the value for the donor atom in germanium is about 34 Å. This is indeed considerably greater than the interatomic distance (which is about 2.44 Å for germanium), hence this picture of the donor center is in this instance to a large extent physically justified.

The ionization energy of the donor center is, from (9.4-4), smaller than that of the hydrogen atom by a factor $1/\kappa^2$ and also by a factor m_n^*/m_0 . For germanium this amounts to $(1/256) \cdot (1/4) = 1/1024$. The energy of ionization for the ground state of a donor atom would then be expected to be about 1000 times less than that of the ground state of the hydrogen atom (13.6 eV). We arrive thus at the figure 0.013 eV for the energy of ionization of a donor atom in germanium. This value is in good agreement with the experimentally determined values obtained from measurements of carrier concentration as a function of temperature and from infrared optical absorption as given in Table 9.1. The higher values of ionization energy observed for silicon are accounted for partially by the smaller dielectric constant ($\kappa = 12$) for

that material, and partly by the higher effective masses for electrons in silicon. The characteristics of acceptor centers may be understood along the same lines of reasoning, using a picture involving a negatively charged acceptor ion and a positively charged hole in a uniform dielectric medium.

TABLE 9.1.
Ionization Energies for Donor and
Acceptor Centers in Ge and Si

Impurity	Ge	Si
P	0.012 eV	0.045 eV
As	0.0127	0.05
Sb	0.010	0.039
B	0.0104	0.045
Al	0.0102	0.06
Ga	0.0108	0.07
In	0.0112	0.16

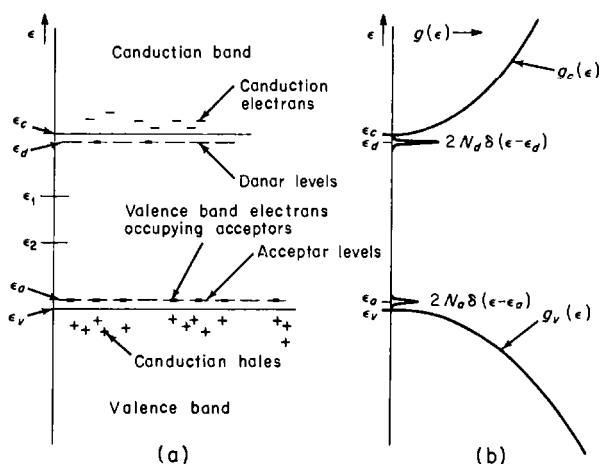


FIGURE 9.8. (a) Energy band diagram of an impurity semiconductor, showing donor and acceptor levels, (b) corresponding density of states curve.

The electrons contributed by donor atoms can thus be construed to originate from localized *donor states* which lie within the forbidden energy gap, a few hundredths of an electron volt *below* the conduction band. Likewise the holes contributed by acceptors can be visualized as being created when electrons which would normally occupy states near the top of the valence band are promoted into initially empty *acceptor levels* which lie in energy a few hundredths of an electron volt above the valence band. This scheme of *donor and acceptor levels* is shown in Figure 9.8(a), and the related density of states picture is shown in Figure 9.8(b).

9.5 STATISTICS OF IMPURITY SEMICONDUCTORS

Before discussing the actual statistical mechanics of impurity semiconductors it is necessary to examine a subtle feature of the occupation statistics of donor and acceptor levels. Confining our discussion for the moment to donor levels, one might be tempted to conclude that the occupation probability associated with such levels would be given simply by the equilibrium Fermi function (5.5-20) with $\epsilon = \epsilon_d$. This is not quite true, however, because of the spin degeneracy of the donor levels. There are really *two* quantum states associated with each donor impurity level, corresponding to the two allowed spin orientations of the electron on the donor atom. But as soon as one of these states is occupied occupancy of the other one is precluded, since the valency requirements of the donor ion are satisfied by one and only one electron. This situation alters the statistical problem which leads to the distribution function.

Consider a system having energy levels of this type, as shown in Figure 9.9. We

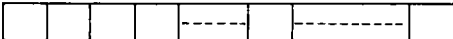
								
Energy level no.	1	2	3	4	---	i	---	n
Energy	ϵ_1	ϵ_2	ϵ_3	ϵ_4	---	ϵ_i	---	ϵ_n
Degeneracy	g_1	g_2	g_3	g_4	---	g_i	---	g_n
No. of electrons	N_1	N_2	N_3	N_4	---	N_i	---	N_n

FIGURE 9.9. Energy levels and notation used in the calculations of Section 9.5.

shall suppose that the electrons occupying the levels are indistinguishable particles, and that there are N_i electrons in the i th level, whose degeneracy is g_i . The number of ways of inserting the first electron into the quantum states belonging to the i th level is g_i ; the number of ways of inserting the second is $g_i - 2$, since the occupancy of the first state by the first electron precludes the occupancy of the state of opposite spin; the number of ways of inserting the third is $g_i - 4$, and so on. For the i th the number will be $g_i - 2N_i + 2$. The total number of ways of arranging N_i indistinguishable electrons in g_i states under these circumstances is then

$$\frac{g_i(g_i - 2)(g_i - 4) \cdots (g_i - 2N_i + 2)}{N_i!} = \frac{2^{g_i/2}(g_i/2)!}{2^{g_i/2 - N_i} N_i! \left(\frac{g_i}{2} - N_i\right)!} \quad (9.5-1)$$

The factor $N_i!$ in the denominator is included because those distributions which are identical except for the permutation of the electrons among themselves must not be counted as distinct distributions, since the electrons are indistinguishable one from another. The total number of ways of realizing a distribution wherein there are $N_1, N_2, \dots, N_n$ electrons in levels 1, 2, $\dots, n$ is just the product of factors of the form (9.5-1) over the levels of the system. This is just the quantity $Q(N_1, N_2, \dots, N_n)$ of Chapter 5 [see, for example, (5.5-3)]. For this system, then, we have

$$Q_d(N_1, N_2, \dots, N_n) = \prod_{i=1}^n \frac{2^{N_i}(g_i/2)!}{N_i! \left(\frac{g_i}{2} - N_i\right)!} \quad (9.5-2)$$

The resulting distribution function is found by maximizing $\ln Q_d$ as before, using the method of Lagrangean multipliers developed in Chapter 5, the result being

$$N_j = \frac{\frac{1}{2}g_j}{1 + \frac{1}{2}e^{(\epsilon_j - \epsilon_f)/kT}}, \quad (9.5-3)$$

where ϵ_f is the Fermi energy, defined in the usual way. For donor levels, the number of donor atoms is just half the number of spin states, or $\frac{1}{2}g_j$, so that for this system, we may write

$$n_d = \frac{N_d}{1 + \frac{1}{2}e^{(\epsilon_d - \epsilon_f)/kT}} \quad (9.5-4)$$

where N_d is the concentration of donor impurity atoms, ϵ_d is the energy of the donor levels, and n_d is the number of electrons per unit volume occupying the donor levels (thus the concentration of *unionized* donors). The Fermi function is clearly modified by the presence of the factor 1/2 before the exponential term in the denominator.

In a generally similar way one may show that if N_a is the concentration of acceptor impurity atoms and ϵ_a the energy of the acceptor levels, then the concentration of holes associated with acceptor atoms (thus the density of unionized acceptor sites) p_a will be

$$p_a = \frac{N_a}{1 + \frac{1}{2}e^{(\epsilon_f - \epsilon_a)/kT}}. \quad (9.5-5)$$

The density of states associated with the donor and acceptor levels may be represented in terms of Dirac δ -functions by

$$g_d(\epsilon) = 2N_d\delta(\epsilon - \epsilon_d) \quad (9.5-6)$$

and

$$g_a(\epsilon) = 2N_a\delta(\epsilon - \epsilon_a). \quad (9.5-7)$$

These equations express the fact that there are no donor or acceptor levels at energies other than ϵ_d and ϵ_a , and that the total number of donor and acceptor quantum states per unit volume is $2N_d$ and $2N_a$, respectively.

If the Boltzmann approximation is valid for *both* the conduction band and the donor levels (that is, if the Fermi level is several kT units below ϵ_d), the exponential factor in the denominator of the expression in (9.5-4) is much greater than unity, and hence the latter may be neglected; Equation (9.5-4) may then be written as

$$n_d = 2N_d e^{-(\epsilon_d - \epsilon_f)/kT}. \quad (9.5-8)$$

Since the concentration of electrons in the conduction band is given by (9.3-6) we may write the ratio of n_d , the number of electrons associated with unionized donors, to $n_0 + n_d$, the total number of free and loosely bound electrons as

$$\frac{n_d}{n_0 + n_d} = \frac{1}{1 + \frac{U_c}{2N_d} e^{-(\epsilon_c - \epsilon_d)/kT}}. \quad (9.5-9)$$

Now $\epsilon_c - \epsilon_d$ is the donor ionization energy, which is ordinarily of the order of or smaller than kT ; the exponential factor above is therefore of the order of unity. If $N_D \ll \frac{1}{2}U_c$ the ratio of the number of electrons on unionized donors to the total number will be very small. Since at 300°K U_c is of the order of 10^{19} cm^{-3} , this condition is satisfied at that temperature for all donor impurity concentrations which are small compared to this figure. In such instances, it is clear that the donors are *almost completely ionized*, and it is usually convenient and accurate to proceed on the assumption that their ionization is complete, setting n_d equal to zero. At very low temperatures the exponential factor in (9.5-9) may be appreciably smaller than unity, and the criterion for complete donor ionization must be written in the more general form

$$N_d \ll \frac{1}{2}U_c e^{-(\epsilon_c - \epsilon_d)/kT}. \quad (9.5-10)$$

In a similar fashion, provided the Boltzmann approximation for holes is valid both in the valence band and at the acceptor levels, one may show that the acceptor levels are practically completely ionized whenever

$$N_a \ll \frac{1}{2}U_v e^{-(\epsilon_a - \epsilon_v)/kT}. \quad (9.5-11)$$

Unless the temperature is very low, this amounts to the requirement $N_a \ll \frac{1}{2}U_v$. In most circumstances, the Fermi level will be within the forbidden gap, many kT units away from both acceptor and donor levels, and both acceptor and donor impurities which may be present will be almost completely ionized.

In a semiconductor at equilibrium there must be either a thermal hole or a positively charged donor ion for every free electron, and a thermal electron or a negatively charged acceptor ion for every free hole. The entire crystal must thus be electrically neutral. This electrical neutrality condition can be expressed by equating the algebraic sum of all negative and positive charges to zero, whereby (recalling that the concentration of ionized donors is $N_d - n_d$ and of ionized acceptors $N_a - p_a$),

$$-e(n_0 + p_0 + N_d - n_d + N_a - p_a) + e(n_0 + p_0 + N_d - n_d + N_a - p_a) = 0. \quad (9.5-12)$$

For all but the lowest temperatures and the highest values of impurity concentration, the Boltzmann approximation will be valid for both conduction and valence bands and for both donor and acceptor levels, whereby the concentrations of unionized donors and acceptors, p_a and n_d , may be neglected in (9.5-12), giving

$$p_0 - n_0 + N_d - N_a = 0 \quad (9.5-13)$$

or, using (9.3-6) and (9.3-15),

$$U_v e^{-(\epsilon_f - \epsilon_v)/kT} - U_c e^{-(\epsilon_c - \epsilon_f)/kT} + (N_d - N_a) = 0. \quad (9.5-14)$$

If we let

$$\alpha = e^{\epsilon_f/kT}, \quad \beta_c = e^{-\epsilon_c/kT} \quad \text{and} \quad \beta_v = e^{\epsilon_v/kT}, \quad (9.5-15)$$

then (9.5-14) can be written in the form

$$\alpha^2 - \frac{N_d - N_a}{U_c \beta_c} \alpha - \frac{U_v \beta_v}{U_c \beta_c} = 0. \quad (9.5-16)$$

Solving this quadratic equation for α , and taking the logarithm of the result, one may obtain

$$\ln \alpha = \frac{\varepsilon_f}{kT} = \ln \left[\frac{N_d - N_a}{2U_c \beta_c} + \sqrt{\left(\frac{N_d - N_a}{2U_c \beta_c} \right)^2 + \frac{U_v \beta_v}{U_c \beta_c}} \right]. \quad (9.5-17)$$

We must choose the positive sign of the radical in (9.5-17) since when $N_d = N_a = 0$, $e^{\varepsilon_f/kT}$ must be a *positive* quantity. This can be put in a somewhat more satisfying form by noting that

$$\ln(a + \sqrt{a^2 + x^2}) = \ln x + \sinh^{-1} \frac{a}{x}. \quad (9.5-18)$$

Using this relation, and substituting the values given by (9.5-15) for β_v and β_c , (9.5-17) can be expressed as

$$\varepsilon_f = \frac{1}{2}(\varepsilon_v + \varepsilon_c) + kT \ln \left(\frac{m_p^*}{m_n^*} \right)^{\frac{1}{2}} + kT \sinh^{-1} \left(\frac{N_d - N_a}{2\sqrt{U_c U_v} e^{-\Delta\varepsilon/2kT}} \right). \quad (9.5-19)$$

The first two terms on the right-hand side of this equation are seen from (9.3-23) to represent the Fermi level ε_{fi} for an intrinsic semiconductor, and since from (9.3-17) and (9.3-20) $n_i = \sqrt{U_c U_v} e^{-\Delta\varepsilon/2kT}$, we may write (9.5-19) finally as

$$\varepsilon_f = \varepsilon_{fi} + kT \sinh^{-1} \left(\frac{N_d - N_a}{2n_i} \right). \quad (9.5-20)$$

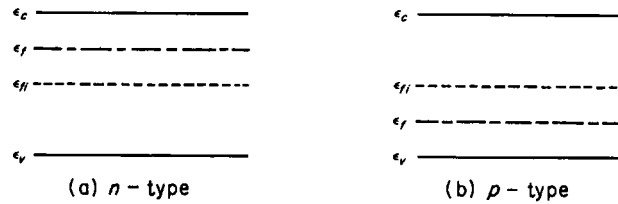


FIGURE 9.10. Relative positions of band edges, intrinsic Fermi level, and actual Fermi level in (a) *n*-type and (b) *p*-type semiconductors.

This expression gives the Fermi level for an impurity semiconductor in the range over which the Boltzmann approximation is satisfied and the donors and acceptors can be regarded as completely ionized. The quantities n_0 , p_0 , n_d and p_a can easily be obtained from (9.3-6), (9.3-15), (9.5-4), and (9.5-5) once the Fermi level has been found.

Since $\sinh^{-1} x$ is positive for $x > 0$ and negative for $x < 0$, it is easily seen that $\varepsilon_f > \varepsilon_{fi}$ for *n*-type semiconductors ($N_d - N_a > 0$), and $\varepsilon_f < \varepsilon_{fi}$ for *p*-type semiconductors ($N_d - N_a < 0$) as illustrated in Figure 9.10. If the number of donors and

acceptors are equal, the inverse hyperbolic sine function vanishes and the material behaves exactly like an intrinsic semiconductor so far as electron and hole populations are concerned. Under these conditions the n - and p -type impurities are said to be fully *compensated*. If the net impurity density $|N_d - N_a|$ is much larger than n_i , the number of thermally excited carriers will be small compared with the total number; in this case the argument of the inverse hyperbolic sine function in (9.5-20) is very large. Since $\sinh^{-1} x \cong \pm \ln |2x|$ for large values of x , it is clear that under these circumstances (9.5-20) becomes

$$\epsilon_f = \epsilon_{fi} \pm kT \ln \frac{|N_d - N_a|}{n_i}, \quad (9.5-21)$$

the plus sign being used for n -type material ($N_d > N_a$), the minus sign for p -type ($N_a > N_d$). A semiconducting material of this type is said to be a strongly *extrinsic* semiconductor, and the range of applicability of (9.5-21) is often referred to as the *extrinsic range*.

The density of carriers in the conduction and valence bands can, of course, be found by substituting the expression (9.5-17) for the Fermi level into (9.3-6) and (9.3-15). These quantities may also be determined from the electrical neutrality condition (9.5-13), with the help of (9.3-20). For example, substituting $p_0 = n_i^2/n_0$ into (9.5-13), we obtain

$$n_0^2 - (N_d - N_a)n_0 - n_i^2 = 0 \quad (9.5-22)$$

which may be solved for n_0 , giving

$$n_0 = \frac{1}{2}(N_d - N_a) + \sqrt{\frac{1}{4}(N_d - N_a)^2 + n_i^2}. \quad (9.5-23)$$

In solving the quadratic equation, it is necessary to choose the positive sign for the radical, since the result must reduce to $+n_i$ for $N_d = N_a = 0$. Likewise, substituting $n_0 = n_i^2/p_0$ into the charge neutrality equation, we can show that

$$p_0 = -\frac{1}{2}(N_d - N_a) + \sqrt{\frac{1}{4}(N_d - N_a)^2 + n_i^2}. \quad (9.5-24)$$

From these formulas, it is clear that when $N_d - N_a = 0$, $n_0 = p_0 = n_i$. This is the case of complete compensation (or, if $N_d = N_a = 0$, of an intrinsic semiconductor). If $N_d - N_a > 0$, $n_0 > p_0$ as expected for an n -type impurity semiconductor; if $N_d - N_a < 0$, $p_0 > n_0$ as required for a p -type semiconductor. For strongly *extrinsic* materials these formulas become much simpler. For example, if $N_d - N_a \gg n_i$, as it is for a strongly n -type sample, then

$$n_0 \cong N_d - N_a \quad \text{and} \quad p_0 \cong \frac{n_i^2}{N_d - N_a} \quad (9.5-25)$$

while for a strongly p -type material where $N_d - N_a$ is large and negative,

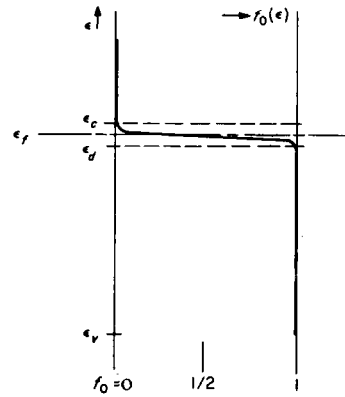
$$p_0 \cong N_a - N_d \quad \text{and} \quad n_0 \cong \frac{n_i^2}{N_a - N_d}. \quad (9.5-26)$$

For the sake of complete generality we have assumed from the beginning that *both* donor and acceptor impurities are present. In many cases, however, especially those involving crystals into which impurities of one type have been intentionally incorporated, the concentration of the *other* impurity species may be completely neglected and indeed set equal to zero in all the formulas of this section.

9.6 CASE OF INCOMPLETE IONIZATION OF IMPURITY LEVELS (VERY LOW TEMPERATURE)

At very low temperatures there may not be sufficient thermal energy available to maintain complete ionization of donor and acceptor impurity levels, as determined by the criteria (9.5-10) and (9.5-11). At zero absolute temperature, for example, in an *n*-type semiconductor, the donor levels must be completely occupied and the conduction band completely devoid of electrons. Both these conditions cannot be realized unless the Fermi level lies *between* the donor levels and the conduction band, as depicted in Figure 9.11. In this case the Boltzmann approximation can no longer be used to

FIGURE 9.11. Distribution function in an *n*-type semiconductor at a very low temperature.



describe the occupation probability associated with the donor levels, since the Fermi level is now above rather than below the donor energy levels. The Boltzmann approximation can, however, still be used for the electrons in the conduction band, because the Fermi level will remain many kT units below the bottom of the conduction band; it should be recalled in this connection that kT is only about 0.001 eV at 10°K.

We shall treat in detail the case of a purely *n*-type semiconductor where $N_a = 0$, and we shall assume that T is so low that the concentration of thermally created holes in the valence band is negligible. Since $N_a = 0$, of course, p_a must also vanish. Setting $N_a = p_a = p_o = 0$ in (9.5-12), one may write

$$n_o - N_d + n_d = 0. \quad (9.6-1)$$

Using (9.3-6) and (9.5-4) to express n_0 and n_d in this equation, we obtain

$$U_c e^{-(\epsilon_c - \epsilon_f)/kT} - N_d \left(1 - \frac{1}{1 + \frac{1}{2} e^{(\epsilon_d - \epsilon_f)/kT}} \right) = 0. \quad (9.6-2)$$

However, since

$$1 - \frac{1}{1 + \frac{1}{2} e^{(\epsilon_d - \epsilon_f)/kT}} = \frac{1}{1 + 2e^{-(\epsilon_d - \epsilon_f)/kT}}, \quad (9.6-3)$$

(9.6-2) can be written as

$$U_c e^{-(\epsilon_c - \epsilon_f)/kT} + 2U_c e^{-(\epsilon_c + \epsilon_d - 2\epsilon_f)/kT} - N_d = 0. \quad (9.6-4)$$

Letting

$$\beta_c = e^{-\epsilon_c/kT}, \quad \beta_d = e^{-\epsilon_d/kT} \quad \text{and} \quad \alpha = e^{\epsilon_f/kT}, \quad (9.6-5)$$

(9.6-4) becomes

$$\alpha^2 + \frac{\alpha}{2\beta_d} - \frac{N_d}{2U_c\beta_c\beta_d} = 0. \quad (9.6-6)$$

This equation can be solved for α to yield

$$\alpha = -\frac{1}{4\beta_d} + \sqrt{\frac{1}{16\beta_d^2} + \frac{N_d}{2U_c\beta_c\beta_d}}. \quad (9.6-7)$$

The plus sign must be chosen in order that $\alpha = e^{\epsilon_f/kT}$ be positive and thus ϵ_f be real. Now, by definition

$$\ln(x + \sqrt{x^2 + a^2}) = \ln a + \sinh^{-1} \frac{x}{a}. \quad (9.6-8)$$

Taking the logarithm of both sides of (9.6-7), using (9.6-8) and reverting to original notation with (9.6-5), one may finally obtain

$$\epsilon_f = \frac{1}{2}(\epsilon_c + \epsilon_d) + \frac{kT}{2} \ln \frac{N_d}{2U_c} - kT \sinh^{-1} \left(\sqrt{\frac{U_c}{8N_d}} e^{-\Delta\epsilon_i/2kT} \right), \quad (9.6-9)$$

where $\Delta\epsilon_i = \epsilon_c - \epsilon_d$ is the donor ionization energy. The density of free electrons in the conduction band may now be calculated from (9.3-6).

At absolute zero, (9.6-9) gives $\epsilon_f = \frac{1}{2}(\epsilon_c + \epsilon_d)$, which means that the Fermi level is midway between the donor levels and the conduction band. As the temperature increases, the Fermi level first increases slightly (remaining nevertheless below ϵ_c) and then decreases, moving down through the donor levels toward the center of the gap. For temperatures such that the Fermi level is several kT units below the donor energy ϵ_d at which point the donors are almost completely ionized, Equation (9.6-9) above

reduces, approximately, to (9.5-21). The proof of this statement is assigned as an exercise. It can be shown in a similar way that for a p -type semiconductor under these same circumstances, the Fermi level at zero temperature lies midway between the acceptor levels and the valence band, and with increasing temperature moves first downward a bit, then upward, approaching the value given by (9.5-21) as the acceptor levels become fully ionized. In both n -type and p -type semiconductors the Fermi level approaches the *intrinsic* value ϵ_{fi} as given by (9.3-23) for sufficiently high temperatures, since at some high temperature the number n_i of thermally excited carriers will be far in excess of the number $|N_d - N_a|$ contributed by donor and acceptor impurities. Beyond this point the behavior of the material will in every way approximate that of an intrinsic semiconductor. In some instances, however, this transition temperature may exceed the melting point of the semiconductor. The Fermi level, as determined by (9.6-9) in the low temperature range and by (9.5-20) at higher temperatures is shown as a function of temperature for both n -type and p -type materials in Figure 9.12.

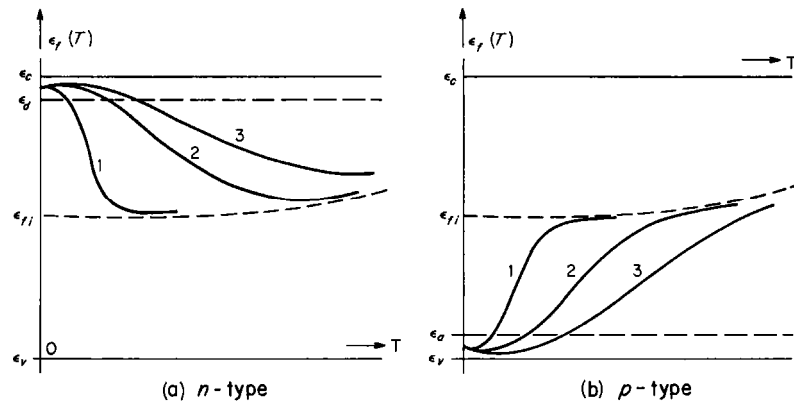


FIGURE 9.12. Variation of Fermi energy with temperature for (a) n -type and (b) p -type semiconductors, of various impurity densities. The curves marked "1" correspond to relatively low impurity densities, those marked "2" to intermediate densities, and those marked "3" to relatively high impurity densities.

9.7 CONDUCTIVITY

The current density carried by holes and by electrons is equal to the product of their respective charge densities and velocities. Thus, the current density, or the charge density per unit time transported by holes and electrons is

$$\mathbf{I}_n = \rho_n \mathbf{v}_n = -en\mathbf{v}_n \quad (9.7-1)$$

and

$$\mathbf{I}_p = \rho_p \mathbf{v}_p = ep\mathbf{v}_p \quad (9.7-2)$$



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